

## SAFETY ASSIGNMENT I: Hazardous Chemical Safety

Attached Files:

-  [HazardousChemicalSafety-Baker.pdf](#)  (1.837 MB)
-  [CHE 382 Safety Quiz.doc](#)  (48.5 KB)

Laboratory safety is an essential part of the Chemical Engineering Laboratory.

1) Read the selected chapters from "**Guidelines on Hazardous Chemical Safety**" from JT Baker Safety Course. (see pdf file)

Chap 3: Toxic chemicals

Chap 8: Flammable chemicals

Chap 9: Corrosive chemicals

Chap 11: Reactive chemicals

Chap 12: Insidious hazards

Chap 13: Compressed gases

2) **Individually complete the safety quiz and submit it to Blackboard by the end of the lab period on January 21st.**

**Submit using the filename: LastName\_CHE382\_SafetyQuiz.doc or LastName\_CHE382\_SafetyQuiz.pdf**

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## SAFETY QUIZ:CHE382-Chemical Engineering Laboratory II

Please answer the following questions testing your knowledge from 6 selected chapters of the JT Baker Hazardous Chemical Safety document; Toxic Chemicals, Flammables, Corrosive Chemicals, Reactive Chemicals, Insidious Hazards and Compressed Gases. Be brief but concise in your answers. Open book/open notes and team discussion is allowed but answers must be in your own words. (5 points per question)

1. What is the general definition of a toxic chemical?

**Any chemical which, when ingested, inhaled, absorbed, injected into or developed within the body, in relatively small amounts, by its chemical action, may cause damage to structure or disturbance to function.**

2. Name the general types of damage that a toxic chemical can cause:

**Local toxicity, systemic toxicity, acute toxicity, chronic toxicity**

3. Name two parameters that can effect toxicity:

**Two from list on page 3-3: Mode of entry, physical condition, dose, duration, sensitivity, combined effects, stress, others (must list)**

4. What is the DOT (Department of Transportation) definition of a nonflammable compressed gas?

**A nonflammable material having a pressure greater than 40 psia at 70°F or 104 psia at 130°F.**

5. a) How does one determine how much material is remaining in a non-liquified gas cylinder?  
**Read the cylinder pressure gauge reading which is proportional to the amount of gas remaining inside the cylinder**

b) On page 13-10, which pressure gauge on the two-stage regulator will give you the cylinder pressure reading?

**The pressure gauge on the left with pressure readings from 500-3000 psi**

6. Other than a single compressed gas, what else can be stored inside a compressed gas cylinder?

**Liquefied gas, a gas dissolved in a liquid or solid, or a compressed gas mixture**

7. Why do insidious hazards represent a problem to workers?

**They represent a problem because the workers may not be aware that a problem exists until chronic, systemic poisoning has occurred.**

8. What are the four general categories of insidious hazards?

**Unanticipated chemical reactions, unanticipated toxicity, improper facility design, improper handling of equipment and chemicals.**

9. Name 2 control measures that can be implemented to reduce insidious hazards in the laboratory.

**See list of 8 control measures on page 12-13**

**Summary of control measures for insidious hazards is given below:**

- 1. Prepare a checklist of insidious hazards. Use the NFPA rating as a guide. A value of 3 or 4 indicates a serious hazard.**
- 2. Thoroughly train all operators in the hazards of substances commonly used in the laboratory.**
- 3. Provide adequate ventilation in the form of hoods and forced air.**
- 4. Do not allow stock build-up of toxic, flammable, or corrosive materials.**
- 5. Have efficient cleanup agents available for spills.**
- 6. Have suitable disaster equipment available (fire extinguishers, gas masks).**
- 7. Secure periodic medical examinations to detect insidious chemical in body tissues.**
- 8. Include an insidious hazards section in your safety audit.**

10. What is the definition of a reactive chemical?

**Reactive chemicals are substances which enter into violent reactions during which the spontaneous liberation of heat and/or gases is too rapid to be safely dissipated by the surroundings.**

11. Name the two groups and their sub-headings used to categorize reactive chemicals:

**Unstable compounds (Explosives, others)**

**Unstable mixtures (Explosives, water-reactive compounds, air-sensitive compounds, other mixtures of oxidizing agents with reducing agents)**

12. List two precautionary measures for handling and/or storing reactive chemicals.

See list on page 11-8 or short paragraph on 11-7.

**11-7:**

**In general, isolate from other potentially reactive compounds. Use adequate protective gear when working with such agents. If the substances involved are not water-sensitive, safety showers, sprinkler systems, and flushing hoses should be made available.**

**11-8: Precautionary Measures:**

**1. Isolate reactive chemicals, store compounds of a class together, away from other classes.**

**2. Have plenty of water available for flushing when water-sensitive chemicals are not involved**

**3. For water-sensitive chemicals, do not allow water in the storage area, or water-type fire extinguishers and sprinklers.**

**4. In general, store reactive chemicals in a cool, dry area, protect from sunlight and label with outdate.**

**5. As a rule, protect reactive chemicals, especially explosives, from shock.**

**6. Store away from flammable and toxic materials**

**7. Wear adequate personal protective equipment. If toxic gases may be generated, self-contained breathing apparatus is warranted.**

**8. Train workers thoroughly in the hazards of the chemicals they are working with.**

**9. Have antidotes, neutralizers, fire-fighting equipment, etc. at hand in case of emergency.**

**10. Have medical assistance available near those areas using reactive chemicals.**

13. How are corrosive chemicals generally defined?

**Corrosive chemicals are substances that, by direct chemical action, are injurious to body tissue or corrosive to metal.**

14. How do corrosive chemicals act on body tissues?

**They act through direct contact with skin or eyes, inhalation or ingestion, and by direct chemical reaction, dissolution of essential components, destruction of protein or disruption of cell membranes.**

15. True or False: Concentrated alkaline solutions have a more corrosive effect on tissue than most strong inorganic acids.

**True**

(Strong alkalies gelatinize tissue, forming soluble compounds and thereby produce deep painful, penetrating burns)

16. Name the six categories of common corrosive liquids.

**Inorganic acids, organic acids, other inorganics, caustic solutions, organic solvents, other organics.**

17. What category of corrosive solids presents the greatest hazard because of their frequent use in industry?

**Caustic alkalies**

18. Name the three components that make up the FIRE TRIANGLE:

**Fuel, oxidizer, ignition source (or energy)**

19. What category of flammables is the most common source of fires in industry?

**Flammable solvents**

20. List two common safety problems and two safety control measures for flammable solvents.

**See lists on page 8-11**

**Common safety Problems Relating to Flammables Solvents:**

**1. Improperly stored glass containers on flammable solvents**

**2. A lack of or inadequate grounding to prevent static sparks, generated by the flowing liquids.**

- 3. Mixed solvents in a waste disposal can**
- 4. A lack of provision for solvent spill cleanup**
- 5. No floor level ventilation.**
- 6. Smoking in solvent areas, ignition sources present**
- 7. Solvents, such as ether, stored in refrigerators which are not explosion proof**
- 8. Inadequate fire extinguishers or automatic extinguisher system**
- 9. Unsafe transport of flammable liquids (glass containers on carts carried in elevators, etc.)**
- 10. Flammable solvents stored with oxidizers or other reactive chemicals.**

#### **Common control measures**

- 1. Isolate solvents from reactive chemicals**
- 2. DO NOT store more than a total of five gallons of solvents in glass in a laboratory.**
- 3. Use an inert atmosphere, if possible, when using solvents**
- 4. Transfer solvents in a working fume hood.**
- 5. Eliminate sources of ignition**
- 6. Store solvents in a cool area, maximum temperature of 80F**
- 7. Transport solvents in metal or other protective containers,**
- 8. Ground all drums and transfer vessels when handling solvents (Grounding clips must contact bare metal)**
- 9. Thoroughly acquaint workers with the hazards of flammable solvents**
- 10. have fire-fighting equipment close at hand AND KNOW HOW TO USE IT**
- 11. Work with and transport solvent only in well-ventilated area.**

#### **BONUS QUESTION:**

Where is the eyeglass cleaning station located in the Unit Operations Laboratory (Room 128)?